

Business Sales, Expense and Profit Trend Analysis **Using Python**

Abstract

Business Sales, Expense and Profit Trend Analysis is an analytics project created in Python that examines a company's financial performance based on two (2) years of monthly sales and expenses data to give the user an idea of how the business is performing for that period.

The data will be processed and analyzed with the use of various Python libraries including Pandas, NumPy, and Matplotlib. The analysis will provide calculations for profit, sales growth percentage, total profit, and maximum profit accompanied by charts depicting the trends in the business.

Problem Statement

Many organizations produce financial data on a continual basis; however, getting meaningful insights from the data can be difficult to obtain. Thus, this project will automate sales and expense analysis and provide accurate calculations and visual representations for businesses in order to produce clear business insights.

Objectives

- Analyze monthly sales and expense data.
- Calculate monthly profit.
- Determine overall business profitability.
- Calculate monthly sales growth percentage.
- Identify maximum profit achieved.
- Visualize business performance through graphs.
- Demonstrate the use of Python in business analytics.

Technologies Used

Programming Language

- Python 3.x

Python Libraries

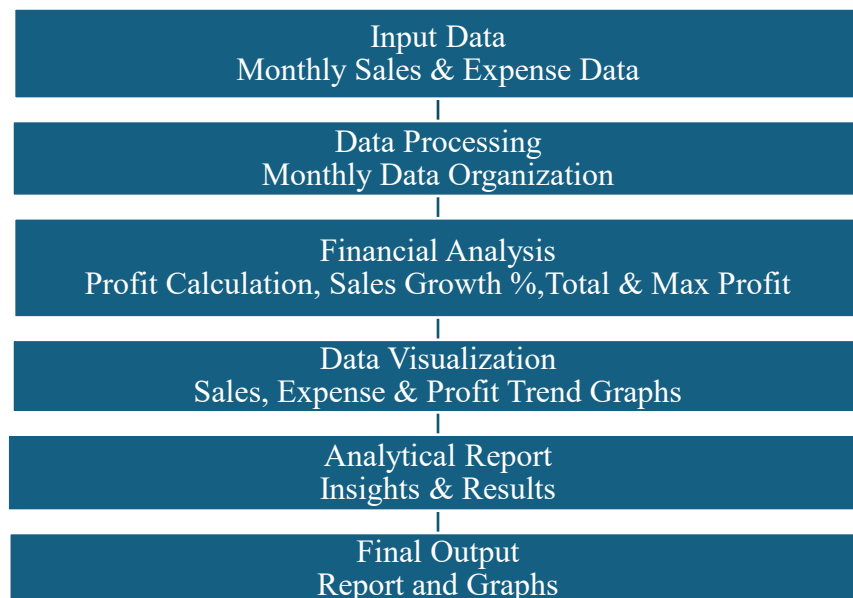
- Pandas

- NumPy
- Matplotlib
- Datetime

Concepts Used

- Variables
- Lists
- DataFrames
- Operators
- Functions
- Conditional Statements
- Loops
- Data Analysis
- Data Visualization

System Flow Diagram



System Design

Input Data

Monthly Sales Data (24 Months)

Monthly Expense Data (24 Months)

Processing

- 1) Create a DataFrame using Pandas.
- 2) Store monthly sales and expenses.
- 3) Calculate monthly profit.
- 4) Calculate sales growth percentage.
- 5) Calculate total profit.
- 6) Calculate maximum profit.
- 7) Generate graphical reports.

Output

- Sales Analysis
- Expense Analysis
- Profit Analysis
- Sales Growth Percentage
- Total Profit
- Maximum Profit
- Business Performance Graph

Algorithm

Step 1: Import required libraries.

Step 2: Create monthly sales and expense datasets.

Step 3: Create a Pandas DataFrame.

Step 4: Calculate profit using: Profit = Sales – Expenses

Step 5: Calculate monthly sales growth percentage.

Step 6: Calculate:

- Average Monthly Growth
- Total Profit
- Maximum Profit

Step 7: Plot Sales, Expenses and Profit trends using Matplotlib.

Step 8: Display analytical results and graph.

Features

- Business performance analysis

- Profit calculation
- Growth trend analysis
- Graphical visualization
- Time-series financial analysis
- DataFrame-based processing
- Easy-to-understand financial reporting

Output

```
months = pd.date_range(start="2023-01-01", periods=24, freq="M")

  Month  Sales  Expenses  Profit
0 2023-01-31  45000     30000   15000
1 2023-02-28  47000     31000   16000
2 2023-03-31  50000     32000   18000
3 2023-04-30  52000     34000   18000
4 2023-05-31  55000     35000   20000

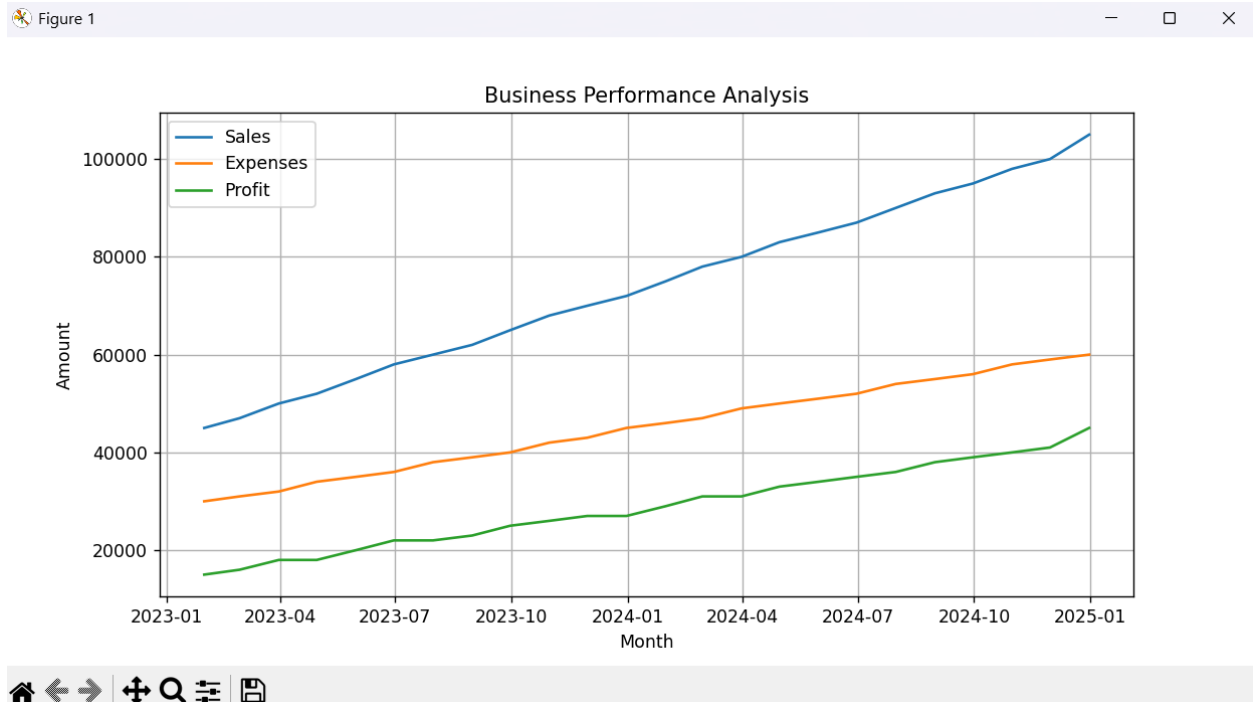
  Month  Sales  Expenses  Profit
19 2024-08-31  93000     55000   38000
20 2024-09-30  95000     56000   39000
21 2024-10-31  98000     58000   40000
22 2024-11-30 100000     59000   41000
23 2024-12-31 105000     60000   45000

Average Monthly Growth
3.7591056114001136

Total Profit
691000

Maximum Profit
45000
```

Graph Analysis



The graph displays three important business indicators:

Sales Trend-Sales show a steady increase over the two-year period, indicating business growth.

Expense Trend-Expenses increase gradually but remain lower than sales.

Profit Trend-Profit remains positive throughout the analysis period, demonstrating sustainable business performance.

Business Insights

- The company-maintained profitability throughout the period.
- Sales growth remained positive.
- Expenses were effectively controlled.
- Profit increased consistently over time.
- Business performance indicates healthy financial growth.

Future Enhancements

- Read data directly from CSV or Excel files.
- Add interactive dashboards.
- Generate automated financial reports.

- Use machine learning for sales forecasting.
- Perform risk and profitability analysis.
- Integrate with real business databases.
- Create a web-based analytics application.

Learning Outcomes

Through this project, I learned:

- Data analysis using Pandas
- Numerical calculations using NumPy
- Data visualization using Matplotlib
- Financial performance evaluation
- Business trend analysis
- Time-series data handling
- Practical application of Python in commerce and finance

Applications

- Financial Reporting
- Business Intelligence
- Management Accounting
- Performance Evaluation
- Sales Forecasting
- Commerce Education
- Data Analytic

Conclusion

The successful Business Sales, Expense and Profit Trend Analysis project illustrates that Python is a viable tool for business and financial analytics. Data analytics and visualization methods used in the project give businesses valuable information regarding their performance and profitability.

This project has shown that today's businesses can leverage analytical tools with Python to aid in decision-making and to develop financial plans.